Software Requirements Specification for SFWRENG 4G06 - Capstone Design Process: DomainX

Team 17, DomainX

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Table 1: Revision History

Date	**Developer(s)**	**Change**
October 10, 2025	Whole team	Initial v0 Draft
December 28, 2025	Fei	Update based on TA and Peer Review feedback issues
March 30, 2026	Fei	Final documentation update contained listed in this Github issue.

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1 Purpose of the Project

The reason for building Domain Comparison Tool can be found in the sections below.

1.1 User Business

The "methodology to assess the state of best practice for Domain X" paper [Smith et al.(October 2021)Smith, Carette, Michalski, Dong, and Owojaiye], outlines a series of systematic steps to assess the state of best practice for a particular domain. However, the process of data gathering is dependent on the student themselves. Current practice involves numerous Excel sheets, and ad-hoc scripts to compile and generate the required tables and graphs. This method is extremely tedious, with lots of manual data entering, and information on prior domains are scattered, with no dedicated central storage. To assist in the creation of this paper, we will create a reusable tool, whose main purpose is to simplify the process of gathering, visualizing and storing the data. This tool will then be reusable for future domains.

1.2 Goals of the Project

The tool will make the data gathering process faster and more efficient for the current paper used to pilot the project. Future papers will then be able to use the tool to assist in their own data gathering for their domain.
See below for our goals, listed in their order of importance.

1.2.1 G1: Centralized Domain Data Location

The tool should be the centralized location where users can view completed data gathered from a domain assessment, as well as create and edit new domains.

1.2.2 G2: Automated Data Collection

The tool must be able to extract the repository based metrics automatically, to eliminate the need for user to manually enter this information. At least 80% of the accessed packages should have these types of metrics automatically filled without error.

1.2.3 G3: Interactive Data Table

The tool's data table should be intuitive to use, and be similar to pre-existing industry standard spreadsheet tools such as Excel. Users with prior experience using Excel should automatically be able to use the tool for inputting data.

1.2.4 G4: Automated Analytic Hierarchy Process

The tool will implement an automated AHP to compare libraries, guiding users through pairwise comparisons to weigh criteria and evaluate alternatives. It should match what the user would calculate exactly if they were to do it manually.

1.2.5 G5: Data Visualization and Download

The tool should allow users to export tables and visuals relating to the domain they want, based on the data gathered in that domain.

1.2.6 G6: Collaboration and Access Control

The tool should allow for multiple users to use the tool simultaneously and contain multiple access levels.

2 Stakeholders

The following section contains the stakeholders of the project.

2.1 Client

The main users for the tool, and the individuals who requested it.

2.1.1 Dr. Spencer Smith

Our supervisor, Dr. Spencer Smith is the product owner of the tool. They are responsible for providing the main features requested, as well has having the final say in the presentation and features of the tool during the team's weekly check-in. When the tool is live, they will also have administrative access and manages the access control of the tool.

Related Goals: On top of the implied goal of having a completed tool, also concerned with G6: Collaboration and Access Control and G1: Centralized Domain Data Location

2.1.2 Research Subteam

The research subteam will be actively using the tool. They are responsible for providing feedback to the product subteam on any pain points experienced while using the tool. As well as assessing if the tool makes the current data gathering process easier, as compared to the previous ad-hoc method of using excel. The research subteam will be alternating between the current students working on this project: Awurama Nyarko, Fei Xie, Ghena Hatoum, and Haniye Hamidizadeh.
Related Goals: G2: Automated Data Collection, G3: Interactive Data Table, G4: Automated Analytic Hierarchy Process, and G5: Data Visualization and Download,

2.2 Customer

The users of the tool.

2.2.1 Research Subteam

See Research Subteam.

2.2.2 Dr. Spencer Smith

See Dr. Spencer Smith.

2.2.3 Ryan Fang

A Master's student at McMaster University currently measuring the State of Best Practices for the Robotics Software domain. Once the tool is in a usable state, after the proof of concept demonstration, he will be onboarded to start using the tool for his research. He is responsible for providing feedback once he starts using the tool.
Related Goals: G2: Automated Data Collection, G3: Interactive Data Table, G4: Automated Analytic Hierarchy Process, and G5: Data Visualization and Download,

2.3 Other Stakeholders

Additional stakeholders for the development or maintenance of the tool.

2.3.1 Development Subteam

The development subteam will be alternating between the current students working on this project. They are responsible for the technical development portion of the project, including testing, system design, developing, and interacting with the client and customers (Dr. Spencer Smith, research subteam and Ryan Fang).
Related Goals: Since the goals are structured from the perspective of the user, there are no explicitly listed goals for the development team, other than the completion of the project.

2.3.2 Dr. Lingyang Chu

The domain expert/consultant for the research subteam during their work on the paper. This person has deep domain knowledge but won't necessarily know the technical software aspects of the domain. He will be interested in the results that the research subteam discovered during their assessment of the domain.
Related Goals: G3: Interactive Data Table, G5: Data Visualization and Download and G6: Collaboration and Access Control

2.4 Hands-On Users of the Project

Broader users of the tool.

2.4.1 Dr. Spencer Smith

See Dr. Spencer Smith.

2.4.2 Research Subteam

See Research Subteam.

2.4.3 Dr. Lingyang Chu

See Dr. Lingyang Chu

2.4.4 Researchers/Industry Workers

These are people who are interested in seeing the results of the research results for a domain. They are responsible for accessing the usability of the tool in the context of a read-only experience, and provide feedback to the product subteam.
Related Goals: G3: Interactive Data Table, G5: Data Visualization and Download and G6: Collaboration and Access Control

2.5 Personas

Below are personas to help visualize the users of the tool.

2.5.1 Kevin - Master's Student at McMaster University

I'm Kevin, a master student at McMaster University currently working on the state of best practice for Domain X. I am extremely busy and don't want to spend too much time entering data. I want a tool to decrease the amount of manual data inputting I need to do, allowing me to focus on the interesting portions of this research. Although I am adequately familiar with software tools, such as using the terminal and version control, my interests are more in the science aspect of computer science andI don't want to spend too long setting things up.

2.5.2 Ashley - Research Scientist at X company

I'm Ashley, a research scientist currently working in the ML/AI field. During my work, I use several software tools to assist me. I don't want to create all the tools myself, preferring to use off-the-shelve, pre-existing libraries, so I can focus on what actually matters. I've read through several tech blogs but they all rank the libraries differently, or don't consider a big enough variety, and sometimes the libraries mentioned are outdated. A consolidated source that can measure varying aspects of the library would be great, since as a researcher, I haven't had as much experience with the DevOps process of the software lifecycle. My personal preference is to use packages and libraries that are easy to learn, install, and use.

2.6 Priorities Assigned to Users

The priority level for each user groups. **Key Users**:

- Research Subteam
- Ryan Fang
- Dr. Spencer Smith

Secondary Users:

- Dr. Lingyang Chu
- Researchers/Industry Workers

2.7 User Participation

The following contains how the development team engages the user during development. **Research Subteam**
Actively using the tool during the writing of the paper. They should provide honest feedbacks on the experience when using the tool. Including any shortfalls, UI disagreements, and bugs.

Domain Expert
Has knowledge of the NN domain, will provide feedback on the research subteam's list of libraries to access, as well as help verify the conclusions during the research process, to see if it is similar from what they have personally experienced. Estimated participation time of: 1-2 hours.

CAS Supervisor
During the user acceptance testing phase of the product. The CASE Supervisor should provide their honest feedback on their experience while using the tool, as they are the main administrator of the tool once it's live.

2.8 Maintenance Users and Service Technicians

Users responsible for maintaining the tool after the initial development **Project Maintainer**
The person responsible for updating the tool with new features, or fixing

any existing bugs once the initial creators of the tool leaves. They were not present during the development of the tool, and will most likely be the future students who use the tool for their domain research.

Infrastructure Team
The infrastructure team at McMaster University will be responsible for hosting the project, and should ensure that the application is available for use and maintains data integrity of their database servers.

3 Mandated Constraints

Constraints on the eventual design of the tool.

3.1 Solution Constraints

Constraints on a particular way the tool should solve and behave.

- Requirement: The product must be built utilizing existing McMaster University software and IT infrastructure.
- Rationale: This approach is mandated to avoid the client incurring costs associated with application and database hosting.
- Fit Criterion: Successful implementation depends on the necessary infrastructure components being formally approved and provided by McMaster University IT services.

3.2 Implementation Environment of the Current System

The current system relies on Excel spreadsheets that are saved and stored either on an individual's personal laptop (which has an unknown operating system) or in an online cloud storage environment.

3.3 Partner or Collaborative Applications

Excel integration is essential. Since Excel was the format utilized by the previous ad hoc tool, enabling planned import and export functionality will

facilitate a smooth transition and allow for the potential backfilling of historical data.

3.4 Off-the-Shelf Software

N/A

3.5 Anticipated Workplace Environment

The application must be fully functional on current and immediately previous stable versions of major web browsers (e.g., Chrome, Firefox, Safari, Edge).

3.6 Schedule Constraints

- Tool must be built by February 2, 2026
- Research paper must be written and submitted by April 6, 2026

3.7 Budget Constraints

The project cannot cost more than $125.

3.8 Enterprise Constraints

N/A

4 Naming Conventions and Terminology

Below are the naming conventions and terminology that will be used throughout the video.

4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

The following terms and acronyms are used throughout this document and among project stakeholders.

NN: Neural Network.

NNL: Neural Network Libraries.

ML: Machine Learning.

AI: Artificial Intelligence.

LLM: Large Language Model.

CAS: Computing and Software Department (McMaster University).

Methodology:
The structured process used to assess the state of best practice within a software domain.

Tool: The software being developed to automate data collection, visualization, and storage.

Research Subteam:
The student group using the tool to apply the methodology and write the research paper.

Domain Expert:
An individual with specialized knowledge in neural networks who provides feedback and validation.

Supervisor:
The CAS faculty member overseeing the project.

State of Best Practice Paper:
The research paper summarizing findings from applying the methodology.

Infrastructure:
University-provided resources (hosting, databases, servers).

Excel Sheets:
Existing manual tools used for data collection.

Ad Hoc Method:
Informal, unstructured approach previously used for data gathering.

Stakeholders:
All individuals involved in or affected by the project (e.g., supervisor, researchers, domain expert).

API : Application Programming Interfaces.

AHP : Analytic Hierarchy Process

BUC : Business Use Cases

5 Relevant Facts And Assumptions

5.1 Relevant Facts

The existing workflow relies on Excel sheets and bash scripts, which do not support automated data gathering.

5.2 Business Rules

Domain experts generally work Monday to Friday, 9 AM – 5 PM, and are unlikely to respond outside these hours.

5.3 Assumptions

McMaster University will provide the infrastructure required to host the application and make the necessary databases available for use.

6 The Scope of the Work

Boundaries of the current environment and business requirements to understand the scope of the product are written below.

6.1 The Current Situation

The following are the steps as stated from the methodology paper [Smith et al.(October 2021)Smith,

1. Domain is identified
2. Research team gathers a preliminary list of packages fitting the domain

3. Domain expert is consulted and helps determine if the packages fit the domain
4. Data gathering commences for the finalized packages used
5. Interviewing or surveying industry users of the packages
6. Research paper is written, with visuals generated from the data

Current ways to consult the domain experts or industry users are done non-systematically, left to the discretion of the research team. The most common method is through email, but this can be difficult to track when multiple messages are sent or several people are being contacted simultaneously.

The data gathering process is performed using Excel spreadsheets, with each entry manually typed. This introduces inefficiencies and potential errors, motivating the need for an automated solution to streamline data collection, improve traceability, and reduce manual workload.

6.2 The Context of the Work

The DomainX Assessment Tool is a web-based application designed to automate the collection, evaluation, and visualization of data for comparing open-source libraries relating to a specific domain. It operationalizes an existing research methodology developed to assess the state of practice in software engineering for a specific domain.

There is currently no automated system performing this type of assessment; existing workflows rely on manual data collection and Excel-based analyses. The tool introduces automation, consistency, and traceability across data gathering, scoring, and visualization activities.

The tool operates within McMaster University's research environment and interacts with multiple adjacent systems for data input, authentication, hosting, and output management. These interactions define the boundaries and environment in which the tool functions.

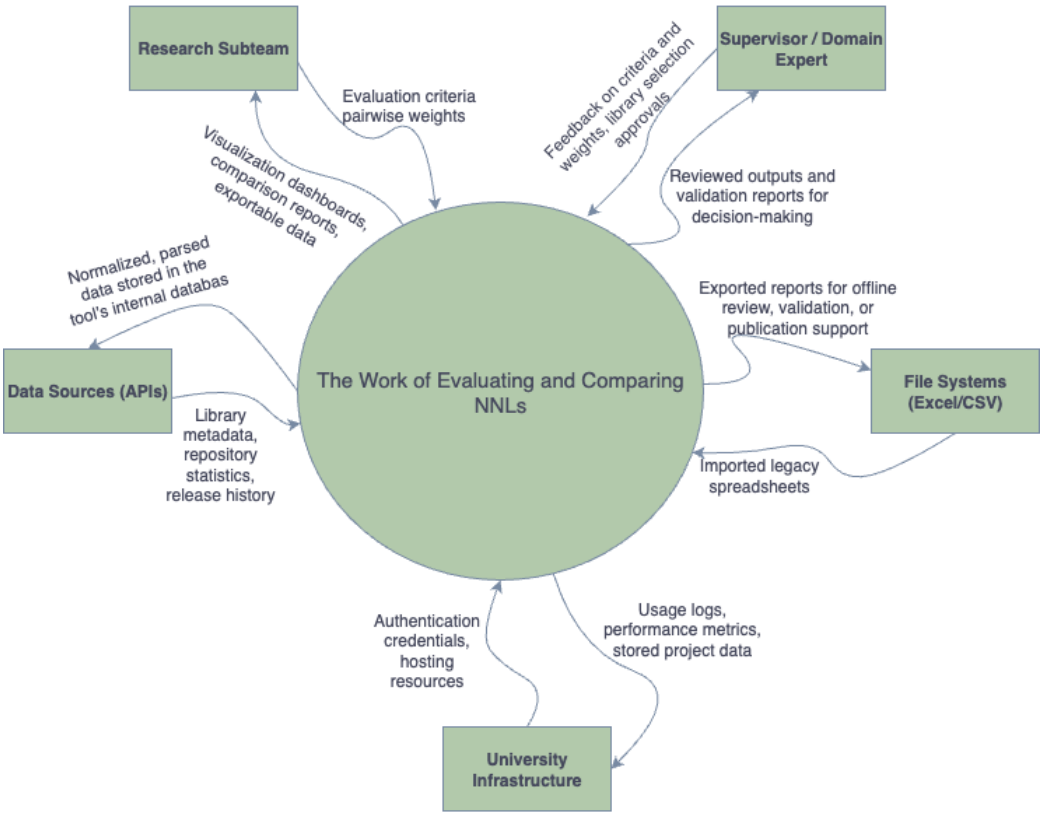


Figure 1: Work Context Diagram for the NNL Assessment Tool

Adjacent Systems:

- **Data Sources (APIs)** – GitHub, PyPI, and official library documentation, providing metadata, repository statistics, and release history.
- **Research Subteam** – Defines evaluation criteria and pairwise weights, initiates analyses, and reviews dashboards.
- **Supervisor / Domain Expert** – Reviews and validates library selections and output reports.
- **University Infrastructure** – Provides authentication services, hosting, and performance monitoring.
- **File Systems (Excel/CSV)** – Enables importing of legacy spreadsheets and exporting reports for offline review or publication.

Each interface between the tool and adjacent systems represents an exchange of data or actions, as shown in Figure 1. Inputs include user-defined criteria and API data; outputs include processed datasets, analytical dashboards, and reports.

All data interfaces will align with the definitions in the Data Dictionary (Section 7).

6.3 Work Partitioning

The following table shows all business events to which the work responds. Business events are happenings in the real world that affect the work. They also happen because it is time for the work to do something.

Table 2: Work Partitioning and Business Use Cases

Event Name	**Input & Output**	**Summary of BUC**
New Library/Package Added for Assessment	**Input:** Research subteam enters GitHub link or package name. **Output:** Tool retrieves metadata (stars, forks, releases, last commit) via API.	System automatically fetches repository metrics from GitHub and stores them in the database for analysis and visualization.
Library Becomes Archived / Deleted / Restricted	**Input:** API request fails or returns error. **Output:** Error log generated; library flagged for manual review.	System detects inaccessible data and notifies research team to investigate and update records.
Domain Expert Revises Library List	**Input:** Feedback from domain expert on selected packages. **Output:** Updated list of libraries for assessment.	System allows easy modification of included libraries; re-runs analysis with updated list.
Assessment Cycle Completed	**Input:** All data collected and validated. **Output:** Generated dashboards, ranking reports, and exportable visualizations.	System computes AHP-based scores and produces comparative visualizations and reports for research publication.

6.4 Specifying a Business Use Case (BUC)

Below is an example of a Business Use Case for a key event in the NNL Assessment Tool workflow.

Table 3: Business Use Case Example – New Package Added to Domain

Business Event	New library/package created for domain usage.
Input(s)	GitHub repository link and metadata retrieved from public API.
Output(s)	Stored library record in database; automated metrics collected (stars, forks, releases).
Trigger	Researcher adds new library or detects a new public repository.
Primary Actor(s)	Researcher.
Secondary Actor(s)	Domain Expert, System APIs.
Preconditions	API access is available; library URL is valid.
Postconditions	Library and metadata stored successfully; flagged for domain expert review.
Main Flow	1. Researcher inputs new repository link. 2. Tool fetches metadata from GitHub and related APIs. 3. Data is validated and stored in the system database. 4. Confirmation and logs generated.
Alternative Flow(s)	– If API call fails, tool logs error and prompts user for manual data entry. – If duplicate record detected, tool alerts user and prevents duplication.
Business Rules	– All repositories must belong to open-source libraries under the defined domain. – Collected metrics must be timestamped for version tracking.
Exceptions	– API rate limits reached. – Network outage during data retrieval.

7 Business Data Model and Data Dictionary

This section contains a specification of the business objects, entities and classes that relate to the product.

7.1 Business Data Model

The following is an entity diagram of the key entities of the product, and how they interact.

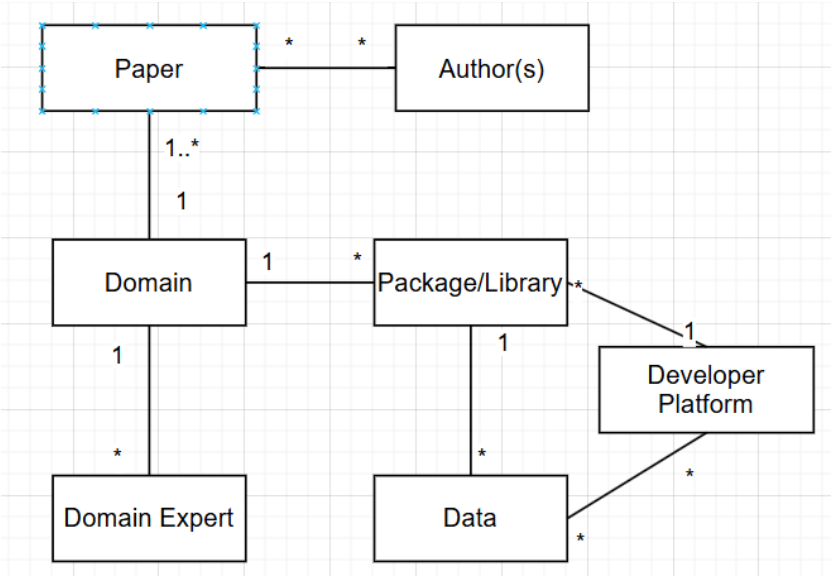


Figure 2: Business Data Model

7.2 Data Dictionary

Name	**Content**	**Type**
Paper	Paper Version + Paper Id + Paper Name	Class
Author/Contributor	Author Name, MacID	Class
Domain	Domain Name	Class
Packages/Library	Package Name + Github URL	Class
Data	Data description, Data Value	Class
Developer Platform	Package URL, Type, API Information	Class
Paper Version	*See Paper*	Attribute/Element
Paper Id	*See Paper*	Attribute/Element
Paper Name	*See Paper*	Attribute/Element
Package Name	*See Package*	Attribute/Element
Package URL	*See Package*	Reference
Data Description	*See Data*	Attribute/Element
Data Value	*See Data*	Attribute/Element
Package URL	*See Developer Platform*	Attribute/Element
Type	*See Developer Platform*	Attribute/Element
API Information	*See Developer Platform*	Attribute/Element

Table 4: Business Data Model Elements

8 The Scope of the Product

Visualization of the scope we decided for the tool.

8.1 Product Boundary

This system aims to simplify and streamline the process of collecting, managing, and analyzing data to assess the state of the practice for various research domains. It replaces the existing, manual process (currently managed via Excel sheets) and formalizes the methodology outlined in the Methodology for Assessing the State of the Practice for Domain X paper. The final system will be a secure, accessible, data management and analysis tool.

The system will provide the following core functionality:

- **Security and User Management:**
  - Securely manage user accounts (login, password changes).
  - Accounts can only be created by an administrator invitation.
  - Enforce Role-Based Access Control (RBAC).
  - Ensure only authenticated users can modify data.
  - Anyone should be able to view available data.
- **Data Integrity and Auditing:** Maintain an audit trail for all data entries, logging when a change is made.
- **Core Data Operations:** Facilitate the storing, updating, and retrieving of all core research entities: research domains, libraries, and collected metrics.
- **Data Collection:**
  - Support mass data upload from previous research work.
  - Automatically collect data to populate a specified data table (e.g., State-of-Practice Metrics Table).
- **Analysis and Visualization:**
  - Implement an AHP-based ranking of libraries within a domain based on their state of practice.
  - Visualize data as different 2D graphs, showing the difference between libraries in a domain.
- **User Experience (UX):**
  - Have an intuitive UI to improve overall user experience.
  - Ensure accessible read permissions for all users.
- **Data Export:** Allow users to download tables or graphs onto their local device.

The following capabilities are explicitly excluded from the initial scope:

- **Communication:** The system will not facilitate scheduling or hosting meetings between researchers and domain experts.

- **User Data:** Only username, email, and password will be collected for user profiles, no need for personal information.
- **Fine-Grained Permissions:** The system will not separate domain access by researcher. All authenticated researchers will be able to view and edit any domain.
- **Data/Library Recommendations:** The system will not give users suggestions on which library to use based on insufficient information.
- **Data Imputation/Handling:** The system will not automatically fill in missing data, nor will it incorporate a mechanism to assume the possibility of missing data during AHP calculation.
- **System Management:** The system will not include functionality for adding or removing administrator accounts.
- **AHP Validation:** The system will not include functionality to check the accuracy of the AHP calculations since external libraries will be used to implement the process itself.

The following items are recognized as potential enhancements but are considered low-priority:

- **Data Auditing:** Logging who edited the data entries.
- **Conflict Resolution:** Creating a method for merging changes when multiple users edit the same data simultaneously.

8.2 Product Use Case Table

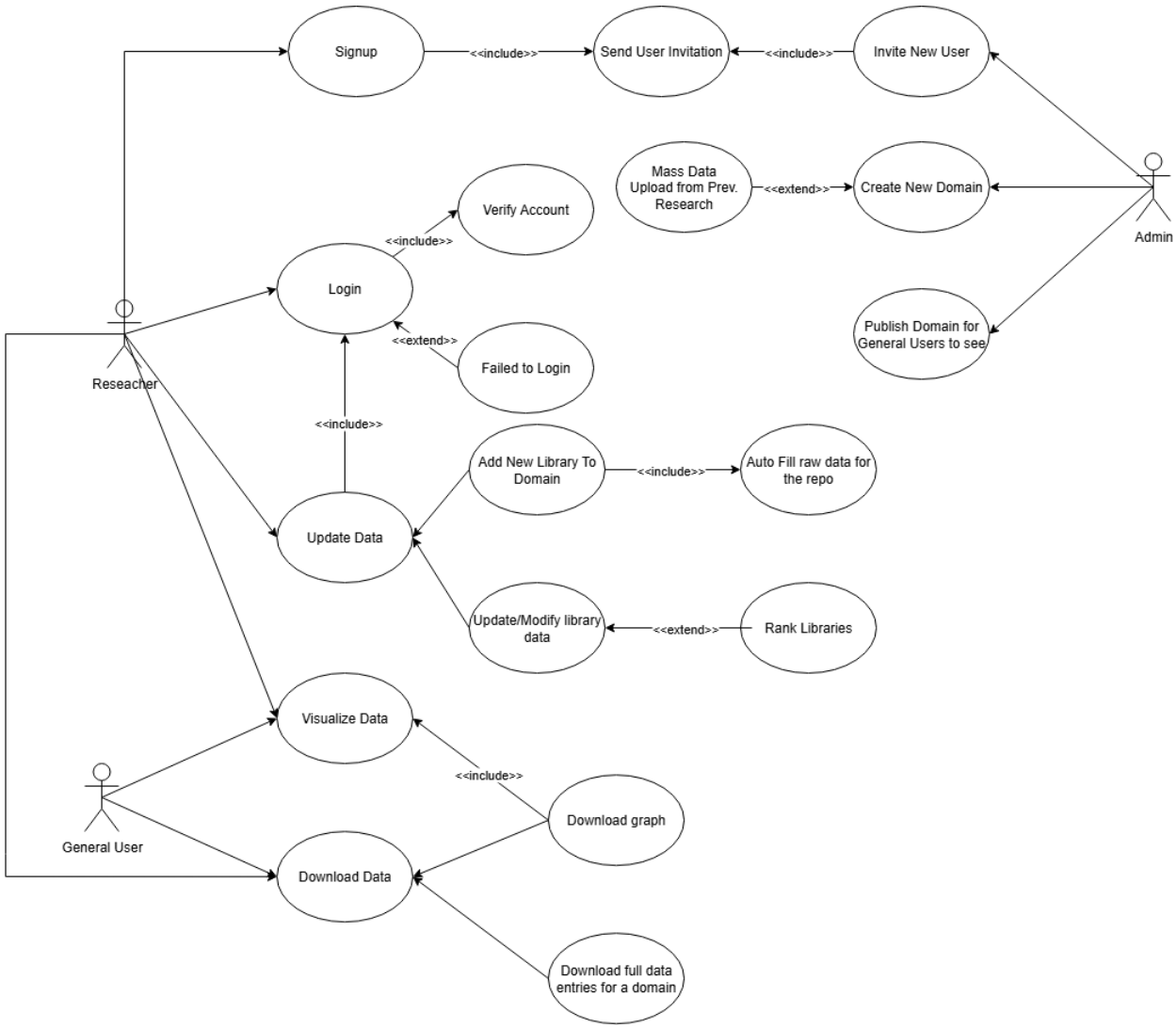
Summary of all the interfaces between the product and other systems.

Use Case (Goal)	Primary Actor	Secondary Actor(s)	Key Functional Requirements (FRs)
Invite New Users	Admin	Researcher	Admin can input a new user email. New user will receive an email with a link that will allow them to setup an account, password, and username.
Domain Creation & Setup	Admin	Researcher	Create new Domain (must validate unique name). Optional: Add description to the domain.
Publish Domain	Researcher	General Users	Once the data collection is finished, allow anyone to see it (General Users).
Automation Library Data	Researcher	System	Auto-fill fields (e.g., creation date, commit date) upon entering a public URL. Check for and flag missing data that was expected to be collected.
Manually Input/Update Data	Researcher	System	Add missing data through the data table. Missing data must be highlighted in red. Only allow user to input right format (e.g., numbers, text). Maintain an audit trail of changes.
Ranking (AHP)	Researcher	System	System will run the integrated AHP tool using input quality scores and expert pairwise comparisons to automatically calculate the final package rankings. Run process automatically when the table is filled and updated.
Visualize & Export Results	General User / Researcher / Admin	System	Allow selection of domain, libraries, and metrics to 2D graphs. Allow download of collected data (JSON/Excel). Allow download of graphs (PNG/LaTeX).

8.3 Individual Product Use Cases (PUC's)

Important things to keep in mind when looking at the Use Case Diagram:

- Include: A include B means, if A is executed that means B will be executed as well.
  - Update Data include Login, if user wanted to Update Data they must Login.
  - Login include Verify Account, when user tries login the system will verify account.
- Extend: A extends B means, if A is executed that means that B could be executed as well but not in all cases.
  - Create New Domain extends Mass Data Upload:Create New Domain could also mean that the user wants to upload previous work, which would need a mass data upload, but not in all cases.
  - Login extends Failed Login: When trying to login the user might fail to do so, however it's not always the case.



9 Functional Requirements

What our tool must do, it's features and behaviours are listed below.

FR1 The Admin will be able to extend an invitation to new researchers via the website, using the researcher's email address.
Rationale: The tool should handle all required functionality for use, without requiring users to go off site.

Fit Criterion: Superuser and admins can send email invite requests from the tool.

FR2 Researchers will be able to sign up using their invited email. The process requires email validation and the creation of a unique username and password.
Rationale: User should be able to create an account through the tool from the invite link.
Fit Criterion: User can create an account and sign up

FR3 The system will allow Admin and Researchers to securely login using their email and password.
Rationale: To allow for role access levels within the application
Fit Criterion: User can sign up with their email and password

FR4 Users will be able to reset their password after validating their email address.
Rationale: In case the user forgets their credentials
Fit Criterion: User receives password reset email if the email exist, and the next login will use the new password instead of the old one to allow user access.

FR5 Admin can create domains, using a unique domain name and optionally a short description.
Rationale: To prevent all users to create domains, only domains that are verified by the product owner (Dr. Spencer Smith) will be able to create domains.
Fit Criterion: Admin users can create domains, while normal users cannot.

FR6 Admin can publish a domain, once research is completed, for anyone to see and download data collected.
Rationale: To allow for domains to be in different stages, and for open access once completed.
Fit Criterion: Unpublished domains cannot be seen by users. Published domains can be seen by users.

FR7 Researchers can add/modify data/libraries of existing domains, and download data/graphs regardless of completion status.

Rationale: Researchers must be able to continuously update and analyze domain data throughout the research lifecycle, including before a domain is finalized or published.
Fit Criterion: Researchers can create, edit, and delete libraries and associated metric data within domains they have access to, and can download datasets and generated graphs whether the domain is published or unpublished.

FR8 The system must validate all user input upon addition or update (e.g., ensuring numeric fields are numbers, dates are valid, and text limits are met).
Rationale: Input validation is required to ensure data correctness, consistency, and reliability for downstream analysis and visualization.
Fit Criterion: Invalid inputs are rejected at submission time with clear error messages, and only data that meets defined validation rules is saved to the system.

FR9 The system must allow an Admin to trigger the automated metric data. The system must report detailed errors for any records that fail validation, such as feature names not being in the scope or wrong data type for a column.
Rationale: Reduces manual data entry effort and minimizes human error when working with multiple libraries.
Fit Criterion: Admins can enter a valid Github URL relating to a library, and the system successfully imports valid metrics while displaying a detailed error report for each invalid row and column.

FR10 When a Researcher enters a public GitHub URL for a library, the system must call the GitHub API to automatically retrieve and populate specific data points, such as the Repository Creation Date and the Last Commit Date.
Rationale: Automatable repository metric data should be automatically extracted.
Fit Criterion: With a valid Github URL, system populates the corresponding rows and columns with the collected metrics, and displays errors for irretrievable data.

FR11 The system must display a complete list of all metrics and their definitions for a selected domain.

Rationale: Users should be able to clearly see what metrics are and what does metrics mean.
Fit Criterion: When a domain is selected, the system displays all associated metrics along with their definitions in a readable format.

FR12 The system will rank the libraries within a domain using the Analytical Hierarchy Process (AHP) and display the calculated results.
Rationale: System should provide a more transparent and automated method to run the AHP between libraries to reduce manual human labour.
Fit Criterion: System computes ranking using AHP and displays final results in a readable and traceable way.

FR13 Users must be able to visualize differences in libraries using comparative graphs.
Rationale: Provides better user readability and quick view of the collected data.
Fit Criterion: Users can select a domain, one or more libraries, relevant metrics, and a graph type, and the system generates the corresponding comparative visualization.

FR14 All users will be able to download the collected data for a specific scope in JSON, Excel, or LaTeX code formats to their device.
Rationale: Providing multiple export formats allows users to reuse data for analysis, reporting, and academic publication.
Fit Criterion: Users can successfully download selected datasets in JSON, Excel, or LaTeX formats, and the downloaded files contain the correct data.

FR15 Users must be able to download any generated graph, with the option to save it as a PNG file.
Rationale: Users may need to include visualizations in reports, presentations, or publications.
Fit Criterion: Users can download any generated graph as a PNG file, and the downloaded image accurately reflects the on-screen visualization.

10 Look and Feel Requirements

Requirements for the appearance of the tool.

10.1 Appearance Requirements

LF-AR1 The application will maintain a consistent visual hierarchy where primary action buttons (e.g., "Save," "Create New Domain") are visually distinct.
Fit Criterion: Primary action buttons use a consistent color and style across the application and are visually more prominent than secondary actions. Users can easily identify buttons within 10 seconds.

LF-AR2 The system must be fully responsive and function correctly on standard desktop monitors and laptops.
Fit Criterion: The application functions correctly at common desktop and laptop screen sizes without overlapping or hidden UI elements.

LF-AR3 Data visualization and editing displays must prioritize information density while remaining readable. Tables should have alternating row colors (zebra striping) to help tracking.
Fit Criterion: Tables with five or more rows use alternating row colors, and all table text remains readable at default browser zoom.

LF-AR4 The main user dashboard should provide a clear, high-level overview of active domains and tasks, with minimal visual clutter.
Fit Criterion: The dashboard displays active domains and tasks on a single screen without requiring scrolling under normal usage.

LF-AR5 Ensure all required input fields are clearly identified (e.g., with an asterisk). Input forms must give real-time feedback, such as a green check mark for acceptable input and a red boundary for errors.
Fit Criterion: All required fields are clearly marked, and validation feedback appears immediately when input is accepted or rejected.

LF-AR6 All error messages (system errors, validation problems) must be clear, short, and contain practical guidance for resolving the problem.
Fit Criterion: Error messages clearly state the issue and suggest a corrective action in one or two short sentences.

10.2 Style Requirements

LF-SR1 The system must use a single, consistent style of icons (e.g., solid, outline, or filled) for all navigational elements, actions, and status indicators.
Fit Criterion: All icons used for navigation, actions, and status indicators follow the same visual style (e.g., all outline or all filled) across the application.

LF-SR2 Graphs must use minimalist design to prioritize data readability. Axes and labels must be clear and legible.
Fit Criterion: All graphs include labeled x- and y-axes with text readable at 100% browser zoom, and labels are legible without zooming on a standard desktop display.

LF-SR3 Font sizing must follow a defined scale (e.g., large for titles, medium for body text, small for captions) and remain consistent on all screens.
Fit Criterion: The application uses no more than three font sizes for titles, body text, and captions, and these sizes are applied consistently across all pages.

11 Usability and Humanity Requirements

Requirements for the usability, and accessibility of the tool.

11.1 Ease of Use Requirements

UH-EU1 The primary navigation menu must be persistently available on the left side or top of every screen, and basic actions (e.g., viewing data, downloading reports) must be achievable within a maximum of three clicks/steps for new users.
Fit Criterion: The primary navigation menu is visible on all screens, and common actions (viewing data, downloading reports) can be completed in three or fewer user actions.

UH-EU2 All data input fields for library metrics must include a tooltip (activated on hover) that explicitly states the required data type and format (e.g., "Numeric value, max 3 decimal places" or "Date in YYYY-MM-DD format").

Fit Criterion: All metric input fields display a tooltip on hover that specifies the expected data type and format.

UH-EU3 All system error messages (validation failures, access denied) must be written in plain language, avoid technical jargon, and provide explicit instructions on how the user can resolve the issue (e.g., "Error: Please verify your username and try again" instead of "401 unauthorized").
Fit Criterion: All system error messages avoid technical error codes and include a brief explanation and a clear corrective action.

11.2 Personalization and Internationalization Requirements

N/A

11.3 Learning Requirements

UH-LR1 All words referring to research entities (Domain, Library, Metric) must be used consistently across the interface and documentation.
Fit Criterion: All references to Domain, Library, and Metric use the same terms across the UI and documentation, with no alternate naming.

11.4 Understandability and Politeness Requirements

UH-UP1 All system-generated error messages will be clear, use non-technical language, and suggest a simple course of corrective action.
Fit Criterion: All system-generated error messages use non-technical language and suggest at least one corrective action.

11.5 Accessibility Requirements

UH-AR1 Critical status indicators and data visualizations must convey meaning through patterns or text labels in addition to color.
Fit Criterion: All critical status indicators and visualizations include text labels or patterns in addition to color.

UH-AR2 Users must be able to change font size using regular browser options (up to 200%) without losing information or functionality.
Fit Criterion: All pages remain readable and functional when browser font size is increased up to 200%.

12 Performance Requirements

Requirements for the performance of the tool itself.

12.1 Speed and Latency Requirements

PR-SL1 The system must finish the AHP ranking calculation for a standard domain in less than ten seconds.
Fit Criterion: The AHP ranking calculation completes in under 10 seconds for a standard domain.

12.2 Safety-Critical Requirements

PR-SC1 The system will encrypt all user passwords.
Fit Criterion: All user passwords are stored in encrypted (hashed) form and never in plaintext.

PR-SC2 If there have been multiple failed attempts to login to an existing account the user must be notified and lock account.
Fit Criterion: After a predefined number of consecutive failed login attempts, the account is locked and the user is notified via email.

PR-SC3 The system will ensure that only researchers modify data.
Fit Criterion: Only users with Researcher or Admin roles can modify domain data. Other users are blocked from write operations.

12.3 Precision or Accuracy Requirements

PR-PA1 All data that the system automatically filled out, must be 100% accurate.
Fit Criterion: All automatically retrieved data matches the API used with transparent processing, if any.

12.4 Robustness or Fault-Tolerance Requirements

PR-RFT1 The system must implement graceful error handling, ensuring that a failure in one function (e.g., a failed GitHub API call) does not crash the entire application or disrupt other users.
Fit Criterion: Failures in external services (e.g., GitHub API) do not crash the application and display a user-facing error message with at least one next step.

PR-RFT2 The system must use database transactions to ensure that data modifications are atomic, if an update operation fails in the middle, the data must be returned to its last validated state.
Fit Criterion: If a data update fails mid-operation, no partial changes are saved to the database.

PR-RFT3 The system should have a protocol in place to take backups of the database, to restore data at max within the timeframe of an hour.
Fit Criterion: Database backups exist and allow restoration of system data within one hour.

12.5 Capacity Requirements

PR-CR1 The system must be able to accommodate up to 20 concurrent active users (logged in and doing operations) with no visible performance decrease.
Fit Criterion: The system supports up to 20 concurrent active users without noticeable performance degradation.

PR-CR2 The database must be able to store at least 70 different study domains.
Fit Criterion: The database can store at least 70 domains without errors.

PR-CR3 The system's architecture must support increasing the underlying database size (up to the maximum capacity defined by the Capacity Requirements) without resulting in database connection errors, timeouts, or data integrity issues.
Fit Criterion: Increasing database size does not cause connection failures, timeouts, or data corruption.

12.6 Scalability or Extensibility Requirements

PR-SE1 The system architecture must clearly divide the Data Management/Storage layer from the Analysis/Visualization layer so that either component may be improved or replaced independently.
Fit Criterion: Data storage and analysis/visualization components are implemented as separate modules.

PR-SE2 The authentication and authorization system must be adaptable enough to accommodate additional user roles (e.g., "Guest Analyst") and specialized rights without requiring code changes to core operations.
Fit Criterion: New user roles can be added through configuration or extension without modifying core application logic.

12.7 Longevity Requirements

PR-LR1 The system must be capable of running without the requirement for maintenance for two years.
Fit Criterion: The system operates without mandatory maintenance interruptions for a period of two years under normal usage.

PR-LR2 Database scaling operations (e.g. migrating to a larger server instance) must be achievable entirely by modifying the system's infrastructure configuration. Such scaling operations must not require any changes to the core application code base.
Fit Criterion: Database scaling can be completed through infrastructure configuration changes without modifying application source code.

13 Operational and Environmental Requirements

Requirements for the operating environment and operational environment the tool will run in.

13.1 Expected Physical Environment

Fit Criterion: The system runs successfully on standard desktop or laptop hardware using a modern web browser and internet connection.

OE-EPE1 The tool is a web-based application that will be used mainly by our team, supervisors, and potentially other researchers or domain experts.

OE-EPE2 It is expected to run on a standard desktop or laptop computer with a reliable internet connection, in a normal indoor setting such as an office, lab, or home workspace.

OE-EPE3 No specialized hardware or rugged equipment is required. A keyboard, mouse, or touchpad, and a modern web browser (e.g., Chrome, Firefox, Edge) are sufficient.

13.2 Wider Environment Requirements

Fit Criterion: The application functions correctly on the latest two to three major versions of Chrome, Firefox, and Edge with an active internet connection.

OE-WE1 The application depends on a stable internet connection for retrieving data from public repositories (e.g., GitHub) and for loading the hosted web interface if deployed to the cloud.

OE-WE2 It does not rely on any dedicated on-premises hardware.

OE-WE3 The tool should work on the latest two to three major versions of common browsers such as Chrome, Firefox, and Edge. No special environmental conditions (lighting, noise, temperature) are expected to affect usability.

13.3 Requirements for Interfacing with Adjacent Systems

The tool must be able to communicate with a few external systems and internal components to collect, store, and visualize data.

OE-IA1 Public Repository APIs (e.g., GitHub API): The tool must communicate with external repository services to automatically collect information about the selected neural-network libraries. For example, it needs to retrieve details such as how often the code is updated, the number of open or closed issues and pull requests, and the main programming

languages used. This data will help evaluate qualities like maintainability, transparency, and overall project activity. The information must be received in a standard machine-readable format (e.g., JSON over HTTPS) whenever a user triggers a scan or during scheduled updates. **Fit Criterion**: System calls external APIs immediately when the scan is triggered, and the system must be able to receive JSON over HTTPS.

OE-IA2 Database (MySQL): The backend must store scores, rankings, and evidence collected from repositories.
Fit Criterion: All scores, rankings, and evidence are stored and retrievable from the database within 10 seconds.

OE-IA3 Visualization Component: The frontend must render charts and comparisons (e.g., using Chart.js) from the data served by the backend.
Fit Criterion: Charts are rendered correctly using backend-provided data using visualization libraries.

OE-IA4 All connections must use standard web technologies (HTTP/HTTPS, JSON) and require only basic authentication methods such as API tokens for secure access to external repository APIs.
Fit Criterion: All external communications use HTTPS and authenticated API access where required.

13.4 Productization Requirements

The tool will be delivered as an open-source web application hosted in the team's public GitHub repository.

The project repository contains a README with setup instructions, required dependency files, and documentation for local and cloud deployment. The application can be accessed via a web URL when deployed, and supports exporting results in standard formats (e.g., CSV, PNG, PDF).

OE-PR1 A clear README file must explain how to set up the backend (Python + Flask) and the frontend (React) using common package managers such as pip and npm.

OE-PR2 For developers running the tool locally, the repository must include a requirements.txt file for Python packages and a database schema file so they can create the required tables.

OE-PR3 When deployed on a cloud platform such as AWS or Google Cloud, users must be able to access the tool directly through a web URL without installing anything.

OE-PR4 The application must also provide options to export results in formats such as CSV for data tables and PNG/PDF for visualizations.

13.5 Release Requirements

Project releases follow the capstone timeline, are published as GitHub Releases with version tags and changelogs, and the final release incorporates documented feedback from earlier demonstrations.

OE-RR1 The project should follow the official capstone timeline: an internal test release before the Proof-of-Concept (PoC) demonstration in **November**, a Revision 0 Demonstration in **Weeks 18–19**, and the Final Release (Revision 1) at **Week 26** along with the research paper and final dataset.

OE-RR2 The Final Release (Revision 1) must incorporate feedback collected during the Revision 0 Demonstration, including usability improvements, bug fixes, and supervisor/TA-requested changes.

OE-RR3 All releases must be published as GitHub Releases and include a short changelog describing the changes in each version.

OE-RR4 Releases must use a clear, descriptive version tag such as the `MAJOR.MINOR.PATCH` format (or an equally descriptive format):

- **MAJOR:** Increased only when a change breaks backward compatibility (e.g., a database schema change that makes older data unusable).
- **MINOR:** Increased when adding new features that remain fully compatible with previous versions.
- **PATCH:** Increased for bug fixes or small improvements that do not affect existing features.

Example version tags:

- `v0.1.0` → first working prototype for the Revision 0 Demonstration
- `v0.2.0` → adds a new feature such as exporting the table to a CSV file
- `v0.2.1` → fixes a small bug in the export feature (patch)
- `v1.0.0` → stable Final Release for Revision 1

14 Maintainability and Support Requirements

Requirements for the maintenance and support of the tool once the initial development has left.

14.1 Maintenance Requirements

The tool is an open-source web application that will need occasional updates to fix bugs and to adapt if external APIs change.

The codebase follows defined coding standards, includes automated formatting and linting tools, maintains test coverage for core functionality, tracks dependencies with pinned versions, logs data changes, and supports routine maintenance and feature updates within the stated timeframes.

MS-MR1 All code must follow the project's coding standards (PEP 8 for the Python backend; React + TypeScript style guide for the frontend).

MS-MR2 Automated tools (such as Black, Flake8, and Pylint) must remain part of the workflow so new contributors can easily read and update the code.

MS-MR3 Unit tests must be written for all new features and bug fixes. Developers should also run basic integration tests to make sure the full pipeline (API → database → visualization) still works after changes.

MS-MR4 Test coverage must be tracked and reported to ensure that critical parts of the backend are being tested.

MS-MR5 All Python and frontend packages must be listed in requirements.txt and package.json, with pinned versions, so the same build can be reproduced.

MS-MR6 The dependency list must be reviewed and updated at least once each semester to keep it current and secure.

MS-MR7 Any changes to the dataset made through the interactive data table must be logged with a timestamp and the username so there is always a clear audit trail.

MS-MR8 Most routine fixes, such as small UI tweaks or bug fixes, should be finished within a couple of days. Larger updates, such as adding a new metric or a new visualization, are expected to take about one to two weeks.

14.2 Supportability Requirements

The tool is intended to be mostly self-supporting since it will be used primarily by our team, the supervisors, and potentially other researchers in the future.

The repository includes up-to-date documentation that allows a new developer to set up and run the system within approximately two hours, provides guidance for core workflows, and includes logging and issue-reporting mechanisms to support troubleshooting.

MS-SR1 The repository must always include an up-to-date guide for installation, setup, the data-collection workflow, using the interactive data table, visualization, and exporting results.

MS-SR2 The backend must include logging to record API failures (such as rate-limit errors or unexpected data formats), database issues, and runtime exceptions so that maintainers can troubleshoot problems quickly.

MS-SR3 Users should use the GitHub Issues page to report bugs or ask questions. No printed manual will be needed; all documentation will remain online in the repository.

14.3 Adaptability Requirements

The system uses a modular, cross-platform architecture that allows new metrics, data sources, user roles, and features to be added with limited changes and without disrupting existing functionality.

MS-AR1 The tool must remain flexible so it can grow with the project and adapt to future needs.

MS-AR2 The MySQL database must be set up so that if a new quality criterion needs to be tracked, it can be added by creating a new column, updating the data-collection script, and adjusting the UI without having to redesign the whole system.

MS-AR3 The data-collection module must allow new data sources (for example, another code-hosting site or a different metrics service) to be added with minimal extra code and without disrupting the existing GitHub integration.

MS-AR4 The tool must be able to run on standard operating systems(Windows, macOS, and Linux) by using widely supported technologies (such as Python, Node.js, and MySQL), and by avoiding any system-specific code.

MS-AR5 The system must be designed in a modular way so that parts like data collection, storage, and visualization remain separate. This makes it easier to add new features or update one part without affecting the rest of the tool.

15 Security Requirements

Requirements on the security in relation to data, such as the privacy and integrity of it.

15.1 Access Requirements

The system enforces role-based access control such that all users can view results, only authenticated Contributors can modify data or perform administrative actions, roles are correctly assigned and enforced, login failures are handled with clear recovery options, and all external API credentials are stored securely outside the source code.

SR-AC1 The tool must be open for anyone to view results, but only approved team members or domain experts may modify the data.

SR-AC2 The system must support two user roles:

- **Viewer:** Can view all interactive data tables that list the libraries, their scores, and rankings, along with all visualizations.
- **Contributor:** Has all Viewer permissions plus the ability to edit data in the interactive table.

SR-AC3 Editing or any other administrative actions must be available only to logged-in users with the **Contributor** role.

SR-AC4 User roles (Viewer / Contributor) must be assigned and updated correctly at sign-up or by an admin so that permissions always reflect the intended access level. *(From FMEA table, Hazard Analysis.)*

SR-AC5 The system must display clear error messages for login failures (e.g., invalid credentials, network errors) and provide a secure way for users to recover or reset their account credentials. *(From FMEA table, Hazard Analysis.)*

SR-AC6 API credentials such as keys or tokens for external services (e.g., repository APIs) must be stored securely outside the source code (e.g., in environment variables) and must never be exposed in the frontend or version control.

15.2 Integrity Requirements

All user inputs are validated before storage, concurrent or conflicting edits are detected and handled without data loss, raw data and derived results are stored separately, publishing is blocked during updates, and users are notified of failed exports or conflicting changes.

SR-INT1 To ensure the accuracy and reliability of the collected data, all manually entered information into the interactive table must be checked before it is saved. For example, numbers must fall within valid ranges, and text must be cleaned so that it cannot be treated as code or scripts.

SR-INT2 The database must include safeguards so that if two users edit the same record at the same time, no changes are lost or overwritten.

SR-INT3 The system must detect simultaneous edits and either block one save or notify the users to resolve the conflict. *(From FMEA table, Hazard Analysis.)*

SR-INT4 When automated data updates conflict with a user's manual edits (if any are updated automatically), the system must warn the user or request confirmation before replacing existing data. *(From FMEA table, Hazard Analysis.)*

SR-INT5 The system must also store the raw evidence, calculated scores, and final rankings in separate fields so that the original data and results cannot be accidentally modified.

SR-INT6 Publishing visualizations must not occur at the same time as data edits or automated refreshes. The system must either block publishing or enforce downtime until updates are complete. *(From FMEA table, Hazard Analysis.)*

SR-INT7 If an export (e.g., CSV, PNG) fails or produces a corrupted file, the tool must alert the user and allow them to retry the download. *(From FMEA table, Hazard Analysis.)*

15.3 Privacy Requirements

The tool mainly works with open-source repository data, so there is very little personal data involved.

Any collected user information is stored securely using standard security practices, and passwords are stored only in hashed form. No additional personal or confidential data is stored by the system.

SR-P1 If basic user details are collected for login (such as name or email), they must be kept private and stored securely using widely accepted security standards.

SR-P2 User passwords must be stored only in hashed form using a secure one-way hashing algorithm so that the actual password is never saved.

No other personal or confidential information will be collected or stored.

15.4 Audit Requirements

The system logs all critical user actions, login events, automated data collection runs, and major errors, and restricts access to these logs to authorized administrators only.

SR-AU1 The system must keep a record of all important activity for accountability and troubleshooting. Every time a Contributor adds, edits, or deletes data in the interactive table, the system must save a log entry showing the user's ID, the time of the action, the type of action, and the fields that were changed.

SR-AU2 The system must also log key events such as login attempts (successful and failed), scheduled API-collection runs, and major errors so that they can be reviewed later.

SR-AU3 Access to these logs must be restricted to authorized project admins only.

15.5 Immunity Requirements

The system prevents common security attacks by using safe database queries, rendering user input as plain text, respecting external API rate limits, and monitoring dependencies for known vulnerabilities.

SR-IM1 The backend must prevent SQL-injection attacks by using the safe query methods provided by the database library instead of building raw SQL strings with user input.

SR-IM2 The frontend must display any text entered by users as plain text only and never allow it to run as code. For example, React's built-in escaping already takes care of this.

SR-IM3 The automated data-collection scripts must respect the rate limits of external repository APIs (e.g., GitHub). If a limit is reached, the scripts should pause and retry later rather than keep sending requests and risk overloading the service.

SR-IM4 All Python and React libraries must be kept up to date, so the project does not rely on outdated packages with known security issues. There should also be a tool in place to regularly check for known vulnerabilities in these libraries, such as GitHub Dependabot.

16 Cultural Requirements

Since the tool will be shared mostly in an academic and research setting, we just need to make sure the interface stays clear, neutral, and easy to understand for anyone who uses it later.

All user-facing text and documentation use plain English, follow consistent date and number formats, apply neutral design choices, and include standard open-source licensing and contribution information.

CU-CR1 **Language:** All text in the user interface and documentation must be in plain English. We should avoid using region-specific jargon so that future collaborators from outside McMaster can understand it easily.

CU-CR2 **Dates and Numbers:** Dates must be displayed in a clear standard format such as ISO 8601 (YYYY-MM-DD) so they're unambiguous for anyone who uses the tool. Numbers must use a consistent style (for example, decimal point for fractions and thousands separated by commas).

CU-CR3 **Neutral Design:** Colours, labels, and icons must stay neutral and must not include any symbols or words that could carry unintended cultural or political meaning.

CU-CR4 **Open-Source Norms:** The repository must include a license notice and simple contribution guidelines to match common open-source practice.

17 Compliance Requirements

The project includes a copyright notice, is distributed under a recognized open-source license, and clearly states citation or acknowledgement requirements for reuse.

17.1 Legal Requirements

CR-LR1 The tool must include a copyright notice covering the code and documentation produced by the team.

CR-LR2 It must be distributed under an appropriate open-source license so that others can reuse it under clearly defined terms.

CR-LR3 Users of the tool or any data generated by it are required to provide proper citation or acknowledgement when they use it in their own work.

17.2 Standards Compliance Requirements

N/A

18 Open Issues

Below is a list of factors and questions that remain unresolved and could significantly influence the design, functionality, or deployment of the Neural Network Libraries (NNL) Assessment Tool.

The purpose of identifying these open issues is to ensure they are tracked, managed, and resolved in a timely manner. Maintaining visibility of these uncertainties supports effective risk management and informed decision-making throughout the development lifecycle.

18.1 List of Open Issues

The below table contains the open issues at the start of the project.

Table 5: Open Issues to Track and Resolve

Issue #	Summary	Cross-Reference	Stakeholders Involved	Action Required	Status
OI-01	Finalization of hosting environment (e.g., internal server)	Section 13 – Operational Requirements	CAS Supervisor, Infra Team	Confirm approved infrastructure with McMaster IT	Pending
OI-02	Confirmation of the final list of Neural Network Libraries	Section 6 – Scope of Work	Research Subteam, Domain Expert	Schedule review meeting with domain expert	Pending
OI-03	Selection of UI framework and visualization library	Section 9 – Functional Requirements	Development Team	Evaluate and finalize tech stack	In Progress
OI-04	Clarification of Excel integration level, if live sync is required or is import/export functionality enough.	Section 9 – Functional Requirements	Research Subteam, Developers	Define required use cases and confirm feasibility	Pending
OI-05	Decision on authentication method (SSO vs custom login)	Section 9 – Functional Requirements	CAS Supervisor, Development Team	Meet with IT to assess SSO feasibility	Pending
OI-06	Confirmation of data storage architecture (SQL vs cloud)	Section 13 – Operational Requirements	Developers, Infra Team	Evaluate hosting options and finalize design	Pending

19 Off-the-Shelf Solutions

This section identifies existing tools, software, and components that could be leveraged to reduce development time and cost for the Neural Network Libraries (NNL) Assessment Tool. It outlines reusable libraries and products that can be legally copied or adapted to accelerate development and ensure maintainability.

The goal is to reuse proven, reliable components and avoid reinventing existing functionality, thereby optimizing development effort and leveraging established best practices.

19.1 Ready-Made Products

Several existing platforms provide partial functionality aligned with the tool's goals, particularly in data visualization, analytics, and automation. However, none fully satisfy the requirement for automated data gathering, Analytic Hierarchy Process (AHP) analysis, and integrated visualization.

Table 6: Ready-Made Products Relevant to the NNL Assessment Tool

Product	**Description**	**Relevance to Project**	**Limitations**
Microsoft Excel	Spreadsheet with storage, formulas, and charts.	Used as baseline for current manual process.	Lacks automation, scalability, and centralized access.
Google Sheets	Cloud-based collaborative spreadsheet.	Enables multi-user editing and sharing.	Limited automation; manual data import.
Power BI / Tableau	Advanced analytics and visualization tools.	Suitable for dashboards and comparative graphs.	Licensing cost; limited AHP customization.
SurveyMonkey / Google Forms	Online data collection tools.	Useful for gathering expert feedback.	No direct database or analytics integration.

These products may serve as inspiration or integration points (e.g., Excel export) but cannot replace the custom automation and analysis required by the NNL Assessment Tool.

19.2 Reusable Components

The following open-source libraries and frameworks will be incorporated to support data collection, analysis, and visualization.

Table 7: Reusable Components

Component	Purpose	Source	Justification
Python Pandas	Data manipulation and analysis.	Open-source	Handles tabular data and transformations efficiently.
Requests (Python)	Retrieve data from GitHub / PyPI APIs.	Open-source	Automates collection of repository metrics.
Matplotlib / Plotly / Chart.js	Visualization libraries.	Open-source	Create interactive and exportable graphs for dashboards.
AHPy / ahpy	Analytic Hierarchy Process implementation.	Open-source	Automates pairwise comparison scoring.
Flask	Backend web framework.	Open-source	Manages API integration and data handling.
React	Frontend user interface framework.	Open-source	Enables responsive dashboards and visualization.
MySQL / SQLite	Database systems.	Open-source	Store collected data and evaluation results.

Reusing these components ensures consistency, leverages reliability, and minimizes custom development effort.

19.3 Products That Can Be Copied

Some open-source or academic research dashboards share functional similarities with the NNL Assessment Tool and may inform design or architecture.

Table 8: Products That Can Be Copied or Adapted

Product	Description	Adaptation Potential
Software Assessment Dashboards	Tools that analyze open-source software metrics.	Their structure can guide data collection, evaluation, and dashboard layout.
University Research Repositories	Academic dashboards for research analytics.	Useful for UI patterns, visualization approaches, and data categorization strategies.

Adaptation saves design time and provides validated frameworks for implementation while maintaining legal compliance.

Considerations:

- Licensing for third-party libraries must be reviewed before adoption.
- Integration testing is required to confirm compatibility.
- Long-term maintenance and documentation quality will influence selection.

Each reused or adapted product will be documented with:

- Name and source
- Functionality
- Integration plan
- Licensing details
- Selection status

20 New Problems

This section identifies potential conflicts, risks, or issues that may arise as a result of implementing the NNL Assessment Tool within McMaster University's research environment. The purpose is to anticipate and document any negative effects or dependencies introduced by the new system.

20.1 Effects on the Current Environment

The new tool will operate within McMaster University's research infrastructure and will rely on institutional hosting (e.g., internal servers). It may introduce additional server load and require IT resources for maintenance.

Motivation: To ensure the new tool integrates smoothly without disrupting existing research tools, storage systems, or academic workflows.

Examples:

- Increased data storage requirements may conflict with current quotas.
- Additional IT workload for managing hosting or user accounts.

Considerations:

- Coordination with the IT infrastructure team is required to confirm compatibility with university servers.
- Ensure the tool does not negatively affect access to existing research applications or networks.

Form: Documented assessment of integration impact with existing systems, supported by infrastructure review.

20.2 Effects on the Installed Systems

The new tool will interface with existing systems such as Excel, university authentication systems (SSO), and potentially university-hosted databases.

Motivation: Identify dependencies between the tool and existing platforms, ensuring stable coexistence and avoiding version conflicts.

Considerations:

- Compatibility with current Microsoft Excel versions.
- Security compliance when connecting to McMaster's SSO.
- Avoid introducing vulnerabilities or version mismatches.

Form: Integration map specifying systems affected, their current versions, and compatibility requirements.

20.3 Potential User Problems

Potential user challenges include onboarding, usability learning curves, and confusion regarding data visualization or interpretation.

Motivation: Ensure researchers and users can adopt the system efficiently without frustration or misinterpretation of outputs.

Considerations:

- Provide user documentation and tutorials.
- Offer training sessions or quick-start guides.
- Establish a support contact for reporting issues.

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Possible constraints include limited hosting capacity, restricted access to certain cloud features, and dependence on McMaster's infrastructure approval.

Motivation: Identify environmental limitations that could delay deployment or reduce tool performance.

Examples:

- Hosting quotas may limit database scaling.
- University IT policies may restrict certain libraries or APIs.
- Limited access to high-performance computing resources.

Considerations: Execute a full review to confirm infrastructure readiness.

20.5 Follow-Up Problems

Potential long-term issues include sustaining the tool after project completion and keeping data updated as new Neural Network Libraries emerge.

Motivation: Ensure the system remains relevant and operational beyond the capstone timeline.

Considerations:

- Define ownership and maintenance responsibilities after project handover.
- Plan for version updates, new library integrations, and user feedback loops.
- Ensure continuity when original developers graduate.

21 Tasks

21.1 Project Planning

The Neural Network Libraries (NNL) Assessment Tool will be delivered using a hybrid development approach that combines Agile iterations with structured milestone-based deliverables aligned with the McMaster Software Capstone schedule.

The lifecycle is divided into major phases: Requirements, Design, Implementation, Testing, and Deployment, each building toward a functional and hosted tool that supports the research team in evaluating neural network libraries.

This approach ensures iterative feedback from supervisors and domain experts after each milestone, enabling continuous refinement. Development will be managed through GitHub for version control, VS Code for coding, and LaTeX for documentation.

The tool will be hosted on McMaster's internal infrastructure or an approved equivalent, with key non-functional activities such as user onboarding, data migration, and training planned in the later stages.

Table 9: Project Planning Phases

Phase	Description	Key Activities	Deliverables
Initiation	Define problem, scope, and objectives.	Document review, team formation, feasibility assessment.	Problem Statement, POC Plan, Development Plan.
Requirements	Gather and formalize system requirements.	Stakeholder interviews, drafting of SRS and Hazard Analysis.	SRS Document, Hazard Analysis.
Design	Establish system architecture and interfaces.	UI mockups, data flow diagrams, schema design.	Design Documents, MG and MIS.
Implementation	Build core tool functionality.	Develop modules, integrate APIs, implement Excel import/export.	Proof of Concept Demo.
Validation & Verification	Test and evaluate system performance.	Unit and integration testing, review sessions, issue resolution.	V&V Plan, V&V Report.
Deployment	Deliver hosted solution.	Deploy tool, prepare documentation, conduct final demo.	Final Hosted Tool, Presentation, Final Reflect and Trace Report.

21.2 Planning of the Development Phases

Each phase contributes to the development of a usable and reliable product. Feedback loops will be incorporated at each stage to ensure alignment with stakeholder expectations, compliance with requirements, and delivery of a secure, maintainable, and user-friendly system.

Table 10: Development Phase Plan

Phase Name	Benefit to User	Operational Date	Operating Components	Functional Requirements	Non-Functional Requirements
Requirements & Analysis	Clarifies system objectives and constraints.	September 5 - September 22	GitHub, LaTeX	Requirements documentation.	Accuracy, clarity.
Design	Defines architecture and interfaces.	September 23 - November 13	UML tools, wire framing software.	UI and backend design.	Maintainability.
Implementation	Delivers core product functionality.	October 27 - February 23	IDE, databases, APIs.	Tool modules, automation scripts.	Reliability, usability.
Testing & Validation	Ensures product meets all criteria.	February 24 - March 27	Test suites, CI/CD.	Verification of features.	Performance.
Deployment	Provides accessible hosted tool.	March 12 - March 27	Hosting platform, documentation.	Hosted application.	Security, accessibility.

22 Costs

There is no development cost for this project, due to the nature of this being a capstone project.
The costs of hosting the required services, such as the database and the web application will depend on McMaster University's existing infrastructure.
However, estimating using a common cloud provider such as Amazon Web Services (AWS).
Using the Relational Database Service (RDS) for database and the Elastic Cloud Computing (EC2) service for hosting. Assuming around a maximum usage of 20 Hours/Month, the total cost of hosting is 11.63 CAD per month,

as shown in Table 11.

Name	Configuration	Estimated Usage	Total Cost
RDS for MySQL	Two vCPU and 8GB of Memory	20 Hours/Month	9.76 CAD/Month
EC2	t4g.large Instance	20 Hours/Month	1.87 USD/Month

Table 11: Price estimates and total costs for two AWS services.

23 User Documentation and Training

23.1 User Documentation Requirements

23.1.1 User Manual

The user manual will highlight the key features of the product, and provide additional details for installing, setup and usage that wasn't previously covered in the project's README.

The maintenance of this document will be the responsibility of the development team. Changes to the product's features, such as adding new features or altering existing features must be reflected in the user manual upon release of the update.

23.1.2 Release Manual

The release manual will cover the release process required for future releases of the product, found in the Development Plan. It should include the whole CI/CD lifecycle, including team standards, release labelling, and more.

The maintenance of this document will be the responsibility of the development team. Changes to the CI/CD process, either from the development team or the broader McMaster University infrastructure team should be reflected before the next release.

23.2 Training Requirements

1. Users should be able to use the tool and utilize key features immediately after following the tutorial

2. Users should be able to find key features without consulting additional documentation 95% of the time.

The responsibility of the training will first fall towards the development team of the tool. They must ensure the provided in-tool tutorial is up-to-date and sufficient to help a user understand all key features.
Additional training will be provided to the supervisor, hosted by the development team. Subsequent training will be the responsibility of the supervisor, if needed to train future users of the tool, in the format that the supervisor chooses.

24 Waiting Room

1. Comparing across Domains
The user should be able to compare two (or more) completed domain analysis against each other.

2. Versioning of Domains
Users can revisit and update completed domains, adding new data or editing existing ones. Allowing users to view the evolution of the state of best practice for the domain.

25 Ideas for Solution

The following is the development team's idea for the user interface of the tool. Figure 3 shows the initial concept, drawing inspiration from Octave Online, the cloud IDE for Matlab.
The main section of the tool will be where the data is displayed and gathered for each domain. With the sections that can be automatically gathered differentiated using a different colour, such as the gray shown in Figure 3.
The left sidebar will contain all the domains, with indications on whether

Figure 3: DomainX UI, inspired by Octave Online

it's completed or not. As well as providing a filter to quickly search for a specific domain.

References

[Smith et al.(October 2021)Smith, Carette, Michalski, Dong, and Owojaiye] Spencer Smith, Jacques Carette, Peter Michalski, Ao Dong, and Olu Owojaiye. Methodology for assessing the state of the practice for domain x. *arXiv preprint arXiv:2110.11575v1*, abs/2110.11575:1–15, October 2021. URL https://arxiv.org/abs/2110.11575.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

Fei

1. **What went well while writing this deliverable?**
   Each team member knew what requirements were, and would work on the sections we decided during the team meeting.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**
   Issues getting the rest of the team to review my work and getting their work up for review. Leaving it hard for me to review their work if it's uploaded so late. This makes it hard to structure PRs, since chances are no one is reviewing my PR even with the team assigned as reviewers, and I can't tell if they seen the comments I left on their PR either. Still wasn't able to completely resolve the issues of reviewing, hopefully a

team meeting specifically on the expectations of reviews will clear this up.

3. **How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?**
   Due to the unique nature of our project, almost all the requirements were inspired by our client, since we are the clients. After reflecting on what we wanted to see when we use the tool in writing the paper, we were able to easily think of requirements.

4. **Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.**
   3RA3 has definitely helped, as well as 4HC3, in terms of understanding what a good layout would be (Users are drawn to colours in the middle of their vision, but more sensitive to motion on the peripherals)

5. **What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.**
   I think the team needs to acquire system design knowledge. Although we've all had experience working adding/updating features of pre-existing projects, starting a completely new one and having to think of all the design requirements will be hard. Additionally, due to the research section, we'll also need to get research, and by expansion, writing knowledge.

6. **For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?**

For my, to pursue the system design knowledge, I believe the best way is to just do it. Early consideration of the design of the project, as well as determining the systems that'll need integrating is crucial. Luckily, we've already started doing that so the rest will just be trial and error. Also reviewing 3RA3 content when needed. For the research/writing knowledge, I'll be reviewing the methodology research paper as well as the existing papers on other domains. As well as consulting with our supervisor during the writing process.

Haniye

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Ghena

1. **What went well while writing this deliverable?** Before meeting the stakeholder, I read over what I sections covered and wrote what my

assumptions were for the requirements. During the meeting, I showed the professor the use case diagram and some of the requirements based on my assumption of how things would work. However, after talking to him there was a few changes that needed to be made, such as it not having to be too secure, allowing all researchers to change any domain, just ensuring that only researchers can modify the data. Also ensured that the stakeholder had an agenda with the questions that we wanted to ask the day before the meeting.

2. **What pain points did you experience during this deliverable, and how did you resolve them?** During our team meeting, we split up the tasks, however we did not realize that we missed some sections until we combined our work. We also didn't review the whole document until the day of summation. My computer wasn't turning on, making it harder to participate in the conversations happening on discord and running make on campus laptops.

3. **How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?** Most of the requirements were inspired from our stakeholder, especially the functional requirements. The security requirements were obtained by the team members, since the stakeholder wasn't thinking too heavily on that section.

4. **Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.** 4HC3 - Human Computer Interfaces, survey stakeholder for to understand how they would like the website to look and how they will be interacting with it, as well as analyze our designs and find their strengths and weaknesses are, 3DB3 - Databases, design a relational database schema based on requirements, 3A04 - Software Design III - Large System Design, analyze what type of design infostructure needed for this project, 2AA4 - Software Design I - Introduction to Software Development, building requirement doc.

5. **What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or soft-**

ware engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member. The team will need to look into the created research paper as well as the steps taken in this area, so reading the Methodology for Assessing the State of the Practice for Domain X paper. We also need to brush-up on design architectures for software systems, to ensure it follows the SOLID principle.

6. **For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?** Read over the Methodology for Assessing the State of the Practice for Domain X paper and start with creating a list of libraries to analyze. Read over previous course notes.

Awurama

1. **What went well while writing this deliverable?**
   I felt like we made good progress connecting the research methodology to the product functional structure. Incorporation of formatted tables and the diagrams and figures as well improved readability and professional presentation. Collaboration through GitHub also went smoothly once we aligned our branch workflow.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**
   The main challenges I faced was resolving Git merge conflicts and maintaining alignment between updated LaTeX sections and binary files like PDFs. This caused delays when pushing new commits. I resolved these issues by communicating with my teammates, cleaning up my local repository, and following a structured branch workflow to avoid future conflicts. Another was formatting the tables to remain consistent across all sections, I resolved this by studying existing examples and researching better ways to make the tables.

3. **How many of your requirements were inspired by speaking to**

your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
Many if not all our requirements are from talking to our "clients" from our project supervisor and our weekly meetings with him or from our team reflections as we are potential users or the researchers that will first utilize the tool.

4. **Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.**
   Probably human computer interfaces as this will help to perfect the user interface and also courses like algorithms and complexity would help with transferrable knowledge to eventually implementing the AHP algorithm.

5. **What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.**
   Our team needs to strengthen technical skills in data integration through APIs, backend development with Flask, and interactive visualization using React or Plotly. We also need to refine project management and documentation practices to ensure smooth collaboration as the project becomes more technical.

6. **For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?**
   For data integration and API automation, one approach is to follow structured tutorials and documentation (e.g., GitHub REST API and Python Requests library). Another is to build small prototypes that

fetch and store data from repositories. I plan to do a combination of learning by doing and also research and learning from tutorials.